

ZIA Traffic Forwarding

Internet Access Issues

	Specific to DC	Specific to Location/Network	Specific to Forwarding method	Specific to User/Group
Primary checks	1. Check Zscaler Trust portal for known issues 2. Check if other locations using same DC are affected	1. Try Telnet on configured proxy IP and port 2. Check if the tunnel is in Up/Down status (tunnel up , line protocol down) 3. Connect Direct to gateway and see if the access is working	1. Test with a different method or manual proxy 2. Check for any recent changes if it was previously working 3. Check reachability to Zscaler gateway	1. Check if Users have common App Profile Or PAC file 2. Check if Users have common routing path or a ODC Firewall 3. Check for a common ZCC version or group policy
Data to Capture	1. MTR or a traceroute to see intermediate path 2. Packet capture while replicating problem 3. Check ZDX dashboard for INcidents with intermediate ISP's	1. Public source IP and tunnel destination IP if any 2. Packet capture while replicating problem 3. ZCC logs if applicable 4. If line protocol is down it indicates issue with monitoring, capture IPSla config details for investigation	1. Details of Forwarding method 2. Packet capture while replicating problem 3. ZCC logs if applicable	1. Details of App Profile/PAC file 2. Weblogs for the time of issue 3. Packet capture while replicating problem 4. ZCC logs if applicable
Basic Analysis	1. Check for local network blocks in capture. Please refer to Appendix 1	1. Check for local network blocks and DNS issues in capture. Please refer to Appendix 2 , Appendix 3 2. Request FW team to validate for block/drops	1. Parse PAC file and find proxy IP/host returned 2. Check local network blocks in capture. Please refer to Appendix 2 3. Request FW team to validate for block/drops	1. Parse PAC file and try Telnet to returned proxy IP or host. 2. Check if there is common group policy with recent changes 3. Check for local network blocks and DNS issues in capture. Please refer to Appendix 2 , Appendix 3
Common Problems	1. Zscaler incident at DC level 2. Problem with the intermediate ISP 3. Problem with customer/Zscaler ISP	1. Network level issue 2. Intermediate ISP issue 3. Proxy misconfig 4. Zscaler cloud issues	1. Network level issue 2. Intermediate ISP issue 3. Proxy misconfig 4. Zscaler cloud issues	1. Network level issue 2. Intermediate ISP issue 3. Proxy misconfig 4. Zscaler cloud issues
Possible Workarounds	Switch to secondary DC. 1. Remove impacted DC from subcloud resolution 2. Fail-over to backup tunnel and/or manually change the primary gateway node in the PAC file	1. By-pass other network devices and connect to internet gateway if that works 2. Switch to fall-back ISP	1. Switch to different forwarding method or an explicit proxy 2. Try a different proxy port if that works	1. Switch Users to working ZCC profile 2. Use default PAC file

Website/Application Access Issues

	Through Zscaler	Specific to DC	Specific to Location/Network	Specific to User/Group
Primary checks	1. Check if the site is accessible without Zscaler . 2. Test with different regions on site24x7 tool	1. Check if the site is accessible without Zscaler. 2. Test with different regions on site24x7 tool 3. Test through secondary DC	1. Check if the site is accessible without Zscaler. 2. Test with different regions on site24x7 tool	1. Check if the site is accessible without Zscaler. 2. Test with different regions on site24x7 tool 3. Check if User has required authorization.
Data to Capture	1. Screenshot of ip.zscaler.com and block page 2. Zscaler weblogs before and after the issue, if it was working previously 3. Header trace if there is specific block message or error code 4. Packet capture if the error is generic	1. Screenshot of ip.zscaler.com and block page 2. Header trace if there is specific block message or error code 3. Packet capture if the error is generic	1. Screenshot of ip.zscaler.com and block page 2. Zscaler weblogs before and after the issue, if it was working previously 3. Header trace if there is specific block message or error code 4. Packet capture if the error is generic 5. ZCC logs if applicable	1. Screenshot of ip.zscaler.com and block page 2. Zscaler weblogs before and after the issue, if it was working previously 3. Header trace if there is specific block message or error code 4. Packet capture if the error is generic 5. ZCC logs if applicable
Basic Analysis	1. Check for SSL errors in capture. Refer to Appendix 4 2. Check weblogs for Zscaler policy action and destination server response code under column "Response Code"	1. Check for SSL errors in capture. Refer to Appendix 4	1. Check for DNS issues. Refer to Appendix 3 2. Check weblogs for Zscaler policy action and destination server response code under column "Response Code"	1. Check if all Users facing issue are pointing to common Zscaler node on ip.zscaler.com 2. Check for DNS issues. Refer to Appendix 3 3. Check weblogs for Zscaler policy action and destination server response code under column "Response Code"
Common Problems	1. Zscaler IP based block on destination server 2. SSL issues or Cookie based auth issues 3. Problems on Zscaler upstream path to the site 4. Proxy based requests blocked by server	1. Zscaler IP based block on destination server 2. Problems on Zscaler upstream path to the server	1. Location or network specific configuration problems 2. DNS issues	1. Location or network specific configuration problems 2. DNS issues 3. Inappropriate changes to ZCC profile or PAC file or Filtering policies applied to impacted users 4. Missing privileges to access the server
Possible Workarounds	1. SSL/Auth Exception 2. Different traffic forwarding method if that works 3. Use SIPA to avoid Zscaler IP based block	1. Route through secondary DC 2. Use SIPA to avoid zscaler IP based block	1. By-pass other network devices and connect to internet gateway if that works 2. Check with fall back ISP	1. Create explicit allow rule if that works 

Performance Issues

	Specific Website/Application	Specific to Location/Network	Single User
Primary checks and Analysis	1. Quantify the difference in the speed with and without Zscaler through packet capture. Refer to Appendix 5 2. Capture header trace and validate page load times with and without Zscaler. Please refer to Appendix 7 3. Check with User/application team if there are any behaviour or protocol differences when there is proxy in the path. 4. Check for any recent change to PAC file based on this host 5. Check for “Proxy Latency” column in the weblogs and see if there is any increase. Proxy latency is the time taken on Zscaler to process the request.	1. Check Zscaler Trust portal 2. Understand the network topology 3. Determine the expected and available throughput. Refer to this article 4. Download a 100MB test file from Azurespeed and monitor Performance under task manager. Refer to Appendix 6 5. Quantify the difference in the speed with and without Zscaler through packet capture. Refer to Appendix 5 6. Test with a different ISP. 7. Check ZDX dashboard to see if ISP has any known incidents. 8. If there is tunnel, by-pass and test with explicit proxy 9. Check and make sure MTU value in case of GRE/IPSec tunnel are optimal. Refer to this article	1. Check for resource utilization stats on TASK manager 2. Determine the expected and available throughput. Refer to this article 3. Download a 100MB test file from Azurespeed and monitor Performance under task manager. Refer to Appendix 6 4. Quantify the difference in the speed with and without Zscaler through packet capture. Refer to Appendix 5 5. Test with a different forwarding mode in case of ZCC.
Information to Capture	1. Screenshot of ip.zscaler.com 2. Enable ZDX probes for impacted site 3. Artifacts to indicate significant difference in speed with and without Zscaler. 4. Packet capture with and without Zscaler while downloading 100MB file. 5. If issue is with any open source application. Share steps with TAC to replicate locally.	1. Traffic forwarding method and network setup details 2. Screenshot of ip.zscaler.com 3. Artifacts to indicate significant difference in speed with and without Zscaler. 4. Packet capture with and without Zscaler while downloading 100MB file. 5. Enable ZDX. 6. MTR trace of 300 pkts to Zscaler IP.	1. Traffic forwarding method. 2. Screenshot of ip.zscaler.com 3. Artifacts to indicate significant difference in speed with and without Zscaler. 4. Packet capture with and without Zscaler while downloading 100MB file. 5. Enable ZDX. 6. MTR trace of 300 pkts to Zscaler IP.
Common Problems	1. Latency/Packet Loss in the routing path between Zscaler and the destination server. 2. Application specific behaviour or protocol difference with and without proxy. 3. Zscaler side processing issues 4. Throttling from destination server 5. Website/Application might be using faster protocols like http2/UDP without Zscaler.	1. Latency/Packet Loss in the routing path between location network and Zscaler DC 2. TCP level configuration issues on the network like sub-optimal MTU, windows scaling issues etc. 3. Location ISP specific issues 4. Utilization issues on the intermediate network devices 5. DNS or PAC file parsing issues 6. Port pool exhaustion on intermediate network device 7. Zscaler side DC / ISP issues	1. Latency/Packet Loss in the routing path between location network and Zscaler DC 2. Connecting to sub-optimal datacenter 3. User’s ISP or intermediate ISP issues. 4. Resource utilization issues on the system. 5. In case of ZCC with DTLS , udp traffic might be blocked by User’s ISP or intermediate ISP. 6. DNS or PAC file parsing issues .
Possible Workarounds	1. Begin with SSL exception to avoid processing overhead , meanwhile TAC guides in identifying root cause and suggests more appropriate workaround	1. Use fallback ISP if the speed is better 2. By-pass any overutilized or network device which is potentially adding latency. 3. Skip DPI if there is latency observed on Firewall	1. Use a different ISP 2. Alter ZCC forwarding mode 3. Switch to simpler PAC file to avoid processing overhead 4. In case of sub-optimal DC, request Zscaler TAC for an IP override to appropriate DC.